

Reviewer Pack — Coxeter Group Enumeration of Boolean Function Entropy via In...

Entry 2eb0cd1151a3 · INCONCLUSIVE

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2026-05-31 00:21:38 UTC

Coxeter Group Enumeration of Boolean Function Entropy via Invariant Generators

Entry ID: 2eb0cd1151a3 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-31 00:21:38 UTC

1. Conjecture

Field A (mathematical branch): Combinatorial Group Theory (specifically, Coxeter groups) **Field B** (complexity object): Boolean function complexity

Statement:

For any boolean function f with n variables and m clauses, the entropy $H(f)$ of f is upper bounded by the product of the number of irreducible generators of the Coxeter group associated with the conflict graph $G(f)$ of f and the logarithm of the maximum size of a coset representative set in the subgroup generated by these generators.

Rationale (proposer's reasoning):

Coxeter groups have been used to analyze combinatorial structures such as graphs, but their potential application to boolean function complexity remains largely unexplored. This conjecture proposes a connection between the structure of Coxeter groups and the inherent randomness or complexity of boolean functions, suggesting that the group's generators could provide insights into the function's entropy.

Taxonomy category: COXETER_GROUP_ENTROPY (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 89682e029c8dbec7

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

For each boolean function f , if the product of the number of irreducible generators of the Coxeter group and the logarithm of the maximum coset representative set size exceeds $H(f)$ by at least a threshold T for all seeds, support the conjecture; otherwise falsify it.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "Coxeter group" AND "Boolean function entropy" - "conflict graph $G(f)$ " AND "irreducible generators" - "subgroup generated by generators" AND "coset representative set" AND "Boolean function complexity"

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_boolean_function(n, m):
        return [random.choice([0, 1]) for _ in range(m)]

    def generate_conflict_graph(f, n):
        G = [[0] * n for _ in range(n)]
        for i in range(len(f)):
            for j in range(i + 1, len(f)):
                if f[i] != f[j]:
                    for k in range(n):
                        if (f[i][k] != f[j][k]):
                            G[k][i] = 1
                            G[k][j] = 1
        return G

    def find_irreducible_generators(G, n):
        generators = []
        for i in range(n):
            if all(G[i][j] == G[j][i] for j in range(i + 1, n)):
                generators.append(i)
        return generators

    def max_coset_representative_set_size(generators, G, n):
        max_size = 0
        for i in range(n):
            size = sum(1 for j in range(n) if all(G[j][k] == G[i][k] for k in generators))
            max_size = max(max_size, size)
        return max_size
```

```

def entropy(f, n):
    ones = f.count(1)
    zeros = len(f) - ones
    p_one = ones / len(f)
    p_zero = zeros / len(f)
    if p_one == 0 or p_zero == 0:
        return 0
    return -p_one * math.log2(p_one) - p_zero * math.log2(p_zero)

results = []
for n in [5, 10, 15, 20, 30, 40]:
    for _ in range(5):
        f = generate_boolean_function(n, random.randint(1, n**2))
        G = generate_conflict_graph(f, n)
        generators = find_irreducible_generators(G, n)
        max_size = max_coset_representative_set_size(generators, G, n)
        H_f = entropy(f, n)
        results.append((n, generators, H_f, max_size))

conjecture_holds = all(len(r[1]) * math.log2(r[3]) >= r[2] for r in results)
counterexample = "" if conjecture_holds else "mapping_undefined"

return {
    "metric_name": "Entropy",
    "metric_value": sum(r[2] for r in results) / len(results),
    "instances_tested": len(results),
    "n_max": max(r[0] for r in results),
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3
    results = []

    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
        results.append(trial_result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std=0.0 support_fraction=1.0")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std=0.0 support_fraction={support_fraction}")
    else:
        first_failing_seed = next(i for i, r in enumerate(results) if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample='mapping_undefined' first_failing_seed={seeds[first_failing_seed]}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3cf1476e.py", line 86, in <module>
    trial_result = run_trial(seed)
                   ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3cf1476e.py", line 62, in run_trial
    G = generate_conflict_graph(f, n)
       ^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3cf1476e.py", line 30, in generate_conflict_graph
    if (f[i][k] != f[j][k]):
        ~~~~^^^
```

TypeError: 'int' object is not subscriptable

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed before producing data, which means it did not complete its execution to verify the conjecture. | next: Debug and fix the error in the test code to allow for a proper verification of the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	19068
2	preregistration	ollama_remote	glm4:latest	0	0	12062
3	novelty	ollama_remote	glm4:latest	0	0	8395
4	novelty	ollama_remote	glm4:latest	0	0	8832
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	15912
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	18552
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12233
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10773
9	judge	ollama_remote	glm4:latest	0	0	11873

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 117698 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/2eb0cd1151a3.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/2eb0cd1151a3.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/2eb0cd1151a3.tar.gz` (if generated)